Exp No:7

Random Sampling and Sampling Distribution

AIM:

To explore random sampling from a population and understand the concept of sampling distribution
using Python in Jupyter Notebook.

Steps:
1. Generate a Population:
○ Create a population of data with a specified distribution (e.g., normal distribution).
2. Random Sampling:
○ Perform random sampling from the population to create multiple samples of different sizes.
○ Compute sample statistics (mean, standard deviation, etc.) for each sample.
3. Sampling Distribution:
○ Plot histograms or density plots of sample statistics (e.g., sample means).
○ Compare the sampling distribution of the sample statistic (mean) with the population distribution.
4. Central Limit Theorem (Optional):
○ Demonstrate the Central Limit Theorem by showing that as sample size increases, the sampling distribution of the sample mean approaches a normal distribution regardless of the population distribution.

PROGRAM:

```
import numpy as np
import matplotlib.pyplot as plt

population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(population_mean, population_std,
population_size)

sample_sizes = [30, 50, 100]
num_samples = 1000

sample_means = {}

for size in sample_sizes:
```

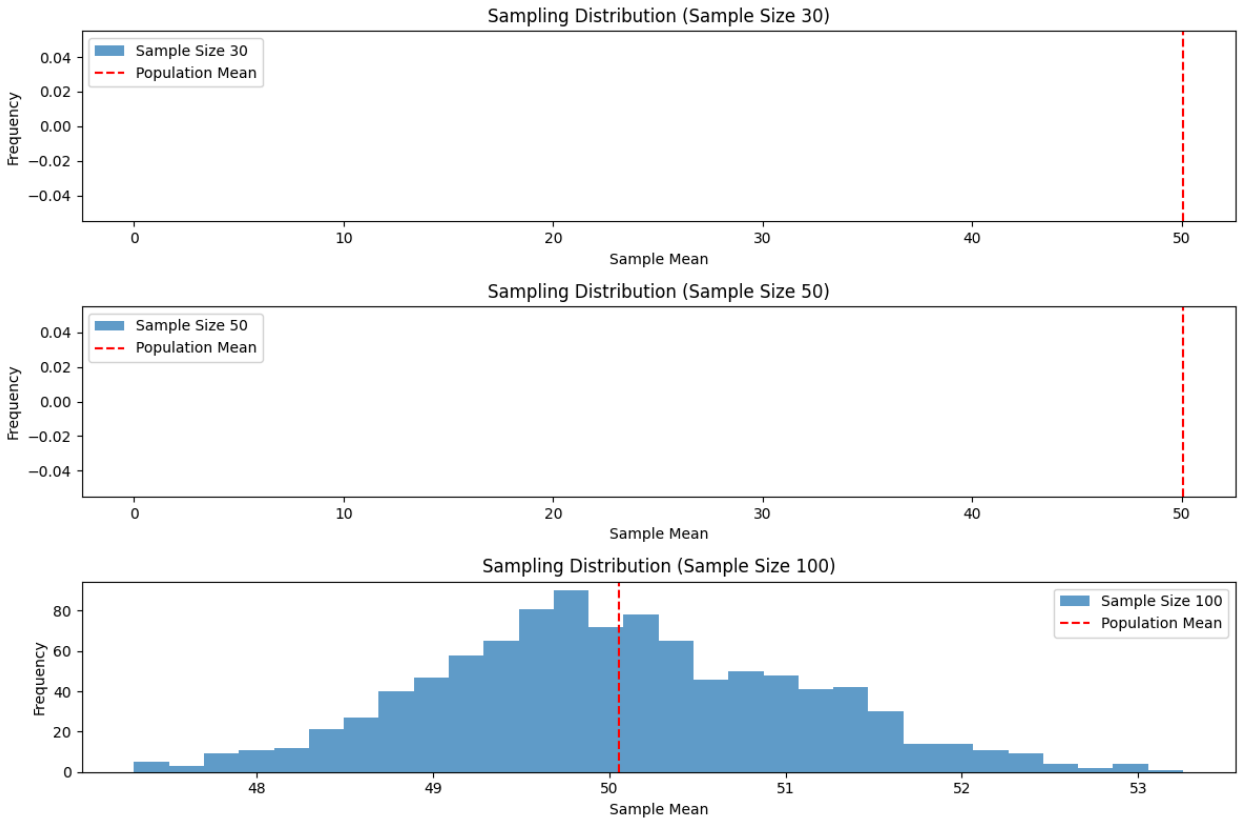
```
  sample_means[size] = []
for _ in range(num_samples):
  sample = np.random.choice(population, size=size, replace=False)
  sample_means[size].append(np.mean(sample))

plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
  plt.subplot(len(sample_sizes), 1, i+1)
  plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size
{size}')
  plt.axvline(np.mean(population), color='red', linestyle='dashed',
linewidth=1.5, label='Population Mean')
  plt.title(f'Sampling Distribution (Sample Size {size})')
  plt.xlabel('Sample Mean')
  plt.ylabel('Frequency')
  plt.legend()

plt.tight_layout()
plt.show()
```

OUTPUT:



RESULT:

The experiment has been carried out successfully